

Mechatronics

Complete student lesson for course code **6021A**.

Revision 2026 Semester 6 Program Elective 4 modules

Course snapshot

Scheme: 4 (L:3, T:1, P:0)

Credits: 4

Contact: 60 periods per semester

Module hours listed: 58

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

How to Study This Lesson

Recommended sequence

1. Begin with the official source declaration and course dashboard.

2. Study each module in the official order and keep the coverage table as a checklist.
3. Open every topic, understand the example or procedure, and write your own short answer.
4. Use the revision section, glossary, questions, and practice paper after completing the modules.

Study principle: Keep English technical terminology, symbols, units, and official assessment wording unchanged in examination answers. Use the Malayalam support blocks only as a bridge to understanding.

Course Dashboard

Course code

6021A

Semester

6

Credits

4

Teaching scheme

4 (L:3, T:1, P:0)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Mechatronics, control systems and sensors	13	CO1
2	Mechanical, fluid-power and electrical actuators	16	CO2
3	PLC architecture and programming	15	CO3
4	Automation and robotics	14	CO4

Prerequisites

- Knowledge of basic Fluid Mechanics and Fluid Power — Fluid Mechanics & Fluid Power, Semester 3; knowledge of basic Electrical & Electronics — Fundamentals of Electrical Engineering, Semester 3.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO-PO Information

Official course objectives

1. To provide knowledge about mechatronics in manufacturing systems and industries.
2. To identify mechanical, hydraulic, pneumatic and electrical actuation systems, and types of sensors and their applications.
3. To gain understanding of digital communications concepts and create PLC programs.
4. To familiarize with principles, characteristics and applications of robotics and automation systems.

Course Outcomes

1. Describe various types of sensors and their applications.
2. Explain mechanical, hydraulic, pneumatic and electrical actuation systems.
3. Explain basic PLC architecture and PLC programming concepts.
4. Describe automation and robotics systems with emphasis on robotic design factors.

CO-PO mapping as supplied

CO1→PO1=2 and PO7=2; CO2→PO1=2 and PO7=2; CO3→PO1=3 and PO7=2; CO4→PO1=3 and PO7=2.

Legend: 3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Importance and applications of Mechatronics	2
module-1	M1.02	Control and measurement systems	3
module-1	M1.03	Types of sensors	4
module-1	M1.04	Sensor selection	1
module-1	M1.05	Sensor and controller applications	3
module-2	M2.01	Mechanical actuators and mechanisms	2
module-2	M2.02	Hydraulic and pneumatic systems and directional valves	3
module-2	M2.03	Pressure and flow-control valves	2
module-2	M2.04	Cylinders and sequencing	2
module-2	M2.05	Process and rotary actuators	2
module-2	M2.06	Electrical actuation systems	3
module-2	M2.07	Stepper motors	1
module-2	M2.08	Basic lift operation	1
module-3	M3.01	PLC definition and block diagram	2
module-3	M3.02	PLC input/output processing and ladder programming	3
module-3	M3.03	PLC mnemonics	2
module-3	M3.04	Timers, relays, counters, shift registers and controls	3
module-3	M3.05	PLC data and analogue I/O	2
module-3	M3.06	PLC and embedded-board applications	3
module-4	M4.01	Automation and robotic systems	2
module-4	M4.02	Robotic applications in manufacturing	3

Module	Topic code	Topic	Hours
module-4	M4.03	Processing, assembly, inspection and feedback	3
module-4	M4.04	Robot programming and design factors	3
module-4	M4.05	Mechatronics case studies	3

Learning Roadmap

1. Mechatronics, control systems and sensors
CO1 · 13 hours

2. Mechanical, fluid-power and electrical actuators
CO2 · 16 hours

3. PLC architecture and programming
CO3 · 15 hours

4. Automation and robotics
CO4 · 14 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Mechatronics, control systems and sensors

This module introduces mechatronics as an integration of mechanical, electrical, electronic, control and computing elements.

Hours: 13

CO1 · Bloom level: Understanding

Learning goals

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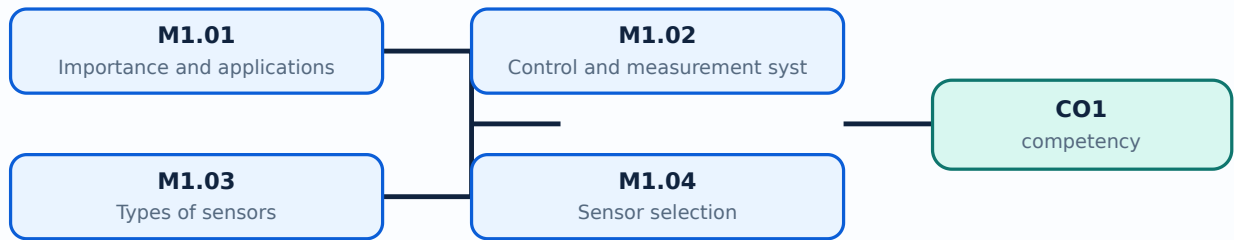
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

Concept and explanation

Official scope. The syllabus specifies **Importance and applications of Mechatronics**. The corresponding study scope is: Explain importance of Mechatronics and its applications in modern industries.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Importance and applications of Mechatronics is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Importance and applications of Mechatronics topic meaning purpose. steps, components variables, application, limitation, safety check Standard technical terms English-Malayalam.

Study points

- Define Importance and applications of Mechatronics using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Importance and applications of Mechatronics is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Importance and applications of Mechatronics under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Importance and applications of Mechatronics without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Control and measurement systems**. The corresponding study scope is: Familiarize control systems and types; open- and closed-loop control; measurement systems and measurement-system terminology.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Control and measurement systems is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Control and measurement systems topic meaning purpose. steps, components variables, application, limitation, safety check Standard technical terms English- Malayalam.

Study points

- Define Control and measurement systems using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Control and measurement systems is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Control and measurement systems under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Control and measurement systems without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Types of sensors**. The corresponding study scope is: Identify different types of sensors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Types of sensors is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Types of sensors topic പഠിക്കേണ്ട വിഷയം meaning പരിഭാഷിക്കേണ്ട purpose പരിശോധിക്കേണ്ട. പരിഭാഷിക്കേണ്ട steps, components പരിശോധിക്കേണ്ട variables, പരിശോധിക്കേണ്ട application, limitation, safety check പരിശോധിക്കേണ്ട പരിശോധിക്കേണ്ട. Standard technical terms English-ലേക്ക് പരിഭാഷിക്കേണ്ട.

Study points

- Define Types of sensors using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Types of sensors is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Types of sensors under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Types of sensors without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Sensor selection**. The corresponding study scope is: Explain selection of sensors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Sensor selection is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** lang="ml">Sensor selection **താഴെ** topic **പരിചയപ്പെടുത്തുക** **പരിചയപ്പെടുത്തുക** meaning **പരിചയപ്പെടുത്തുക** purpose **പരിചയപ്പെടുത്തുക**. **പരിചയപ്പെടുത്തുക** **പരിചയപ്പെടുത്തുക** steps, components **പരിചയപ്പെടുത്തുക** variables, **പരിചയപ്പെടുത്തുക** application, limitation, safety check **പരിചയപ്പെടുത്തുക** **പരിചയപ്പെടുത്തുക** **പരിചയപ്പെടുത്തുക**. Standard technical terms English-**മലയാളം** **പരിചയപ്പെടുത്തുക**.

Study points

- Define Sensor selection using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Sensor selection is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Sensor selection under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Sensor selection without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Sensor and controller applications**. The corresponding study scope is: Identify sensor and controller applications in washing machines and automatic water-level controllers.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Sensor and controller applications is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:**
 Sensor and controller applications topic meaning purpose. steps, components variables, application, limitation, safety check Standard technical terms English-

Study points

- Define Sensor and controller applications using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Sensor and controller applications is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Sensor and controller applications under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Sensor and controller applications without explaining their order, purpose, or engineering consequence.

Module summary: Sensor selection should match range, sensitivity, response, environment, interface and required accuracy.

OFFICIAL MODULE 2

Mechanical, fluid-power and electrical actuators

This module explains how control signals become mechanical motion or force through mechanical, hydraulic, pneumatic and electrical actuators.

Hours: 16

CO2 · Bloom level: Understanding

Learning goals

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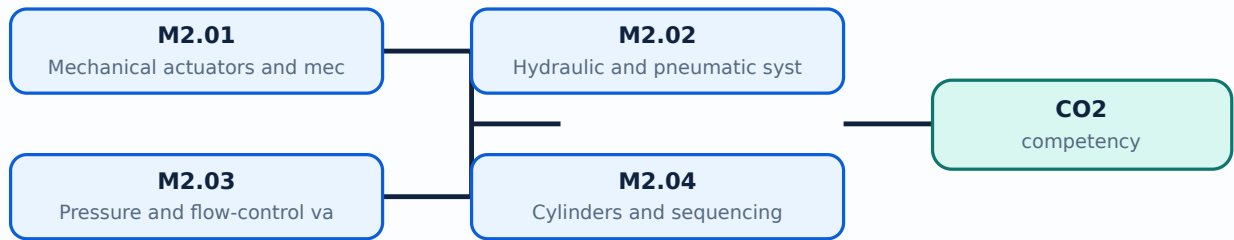
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

Engineering / workplace application: Mechanical actuators and mechanisms is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Mechanical actuators and mechanisms under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Mechanical actuators and mechanisms without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Hydraulic and pneumatic systems and directional valves**. The corresponding study scope is: Familiarize hydraulic and pneumatic systems, power-supply layout, spool/poppet/pilot-operated and directional control valves.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Hydraulic and pneumatic systems and directional valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: Malayalam support: Hydraulic and pneumatic systems and directional valves topic meaning purpose. steps, components variables, application, limitation, safety check. Standard technical terms English-Malayalam.

Study points

- Define Hydraulic and pneumatic systems and directional valves using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.

- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Hydraulic and pneumatic systems and directional valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Hydraulic and pneumatic systems and directional valves under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Hydraulic and pneumatic systems and directional valves without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Pressure and flow-control valves**. The corresponding study scope is: Familiarize pressure-regulating, pressure-limiting, pressure-sequence and flow-control valves.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Pressure and flow-control valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Pressure and flow-control valves topic meaning purpose. steps, components variables, application, limitation, safety check Standard technical terms English- Malayalam.

Study points

- Define Pressure and flow-control valves using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Pressure and flow-control valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Pressure and flow-control valves under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Pressure and flow-control valves without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Cylinders and sequencing**. The corresponding study scope is: Identify single-acting and double-acting cylinders and cylinder sequencing.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Cylinders and sequencing is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Cylinders and sequencing topic പദപരിചയം meaning പദപരിചയം purpose പദപരിചയം. പദപരിചയം steps, components പദപരിചയം variables, പദപരിചയം application, limitation, safety check പദപരിചയം പദപരിചയം പദപരിചയം. Standard technical terms English-പദപരിചയം പദപരിചയം.

Study points

- Define Cylinders and sequencing using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Cylinders and sequencing is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Cylinders and sequencing under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Cylinders and sequencing without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Process and rotary actuators**. The corresponding study scope is: Familiarize process-control valve, diaphragm actuator, rotary actuator and semi-rotary actuator.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Process and rotary actuators is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Process and rotary actuators
topic
meaning
purpose
steps, components
variables,
application, limitation, safety check
Standard technical terms English-

Study points

- Define Process and rotary actuators using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Process and rotary actuators is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Process and rotary actuators under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Process and rotary actuators without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Electrical actuation systems**. The corresponding study scope is: Describe mechanical switches, relays, solid-state switches, diodes, thyristors, triacs, bipolar transistors and solenoids.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Electrical actuation systems is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Electrical actuation systems
topic
meaning
purpose
steps, components
variables,
application, limitation, safety check
Standard technical terms English-

Study points

- Define Electrical actuation systems using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Electrical actuation systems is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Electrical actuation systems under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Electrical actuation systems without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Stepper motors**. The corresponding study scope is: Explain the working principle of stepper motors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Stepper motors is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: Malayalam support: lang="ml">Stepper motors എന്ന തീർച്ചസ്ഥിതം വിവരിക്കുക. ഇതിന്റെ അർത്ഥം, പ്രവർത്തന രീതി, ഘടകങ്ങൾ, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന എന്നിവ വിശദീകരിക്കുക. ഔദ്യോഗിക സാങ്കേതിക പദങ്ങൾ ഉപയോഗിക്കുക.

Study points

- Define Stepper motors using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Stepper motors is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Stepper motors under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Stepper motors without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Basic lift operation**. The corresponding study scope is: Discuss the application of basic lift operation.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Basic lift operation is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Basic lift operation വിഷയം പഠിക്കേണ്ട വിഷയം meaning പഠിക്കേണ്ട വിഷയം purpose പഠിക്കേണ്ട വിഷയം. പഠിക്കേണ്ട വിഷയം steps, components പഠിക്കേണ്ട വിഷയം variables, പഠിക്കേണ്ട application, limitation, safety check പഠിക്കേണ്ട വിഷയം പഠിക്കേണ്ട. Standard technical terms English-ൽ പഠിക്കേണ്ട വിഷയം പഠിക്കേണ്ട.

Study points

- Define Basic lift operation using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Basic lift operation is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Basic lift operation under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Basic lift operation without explaining their order, purpose, or engineering consequence.

Module summary: An actuator answer should identify energy source, control element, motion/force output and feedback or safety limitation.

OFFICIAL MODULE 3

PLC architecture and programming

This module develops PLC architecture, ladder logic, mnemonics, timing, counting, data handling and small embedded-board applications.

Hours: 15

CO3 · Bloom level: Applying

Learning goals

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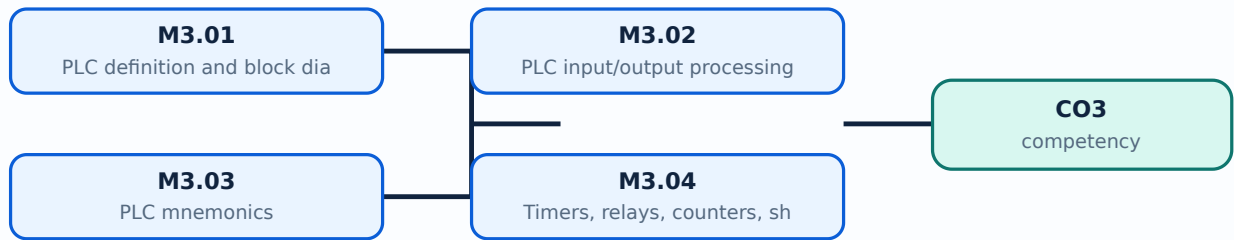
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

Concept and explanation

Official scope. The syllabus specifies **PLC definition and block diagram**. The corresponding study scope is: Explain Programmable Logic Controller definition and basic block diagram.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC definition and block diagram is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** lang="ml">PLC definition and block diagram topic പഠിക്കേണ്ട വിഷയം PLC ന്റെ അർത്ഥം പരിശോധിക്കുക പരിശോധിക്കേണ്ട ലക്ഷ്യം PLC ന്റെ പ്രവർത്തനം, ഘടകങ്ങൾ, വേരിഫിക്കേഷൻ, പരിമിതി, സുരക്ഷാ പരിശോധന, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന, പരിശോധന. Standard technical terms English-മലയാളം പദപരിവർത്തനം.

Study points

- Define PLC definition and block diagram using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: PLC definition and block diagram is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for PLC definition and block diagram under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for PLC definition and block diagram without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies PLC input/output processing and ladder programming. The corresponding study scope is: Discuss input/output processing; ladder diagram, logic functions, latching and sequencing.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write input → transformation → output → verification. For a design or management method, write objective → assumptions → method → result → limitation.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC input/output processing and ladder programming is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: PLC input/output processing and ladder programming എന്നു് വിഷയം പഠിക്കുന്നതിനുള്ള പഠനരീതിയും അതിന്റെ അർത്ഥവും, പ്രവർത്തനവും, ഘടകങ്ങളും, പ്രയോഗവും, പരിമിതിയും, സുരക്ഷാ പരിശോധനയും എന്നിവയെക്കുറിച്ചും. Standard technical terms English-ലേക്ക് മാറ്റി എഴുതണം.

Study points

- Define PLC input/output processing and ladder programming using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: PLC input/output processing and ladder programming is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for PLC input/output processing and ladder programming under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for PLC input/output processing and ladder programming without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **PLC mnemonics**. The corresponding study scope is: Familiarize PLC mnemonics. **How to study the topic.** Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC mnemonics is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** PLC mnemonics എന്ന വിഷയം പരിചയപ്പെടുക. PLC എന്നതിന്റെ അർത്ഥം, പ്രവർത്തനം, ഘടകങ്ങൾ, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന എന്നിവയെക്കുറിച്ച് വിശദീകരിക്കുക. പരമ്പരാഗത സാങ്കേതിക പദങ്ങൾ ഇംഗ്ലീഷിൽ നിന്ന് മലയാളത്തിലേക്ക് തിരിയെഴുതുക.

Study points

- Define PLC mnemonics using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: PLC mnemonics is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for PLC mnemonics under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for PLC mnemonics without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Timers, relays, counters, shift registers and controls**. The corresponding study scope is: Discuss timers, internal relays, counters, shift registers, master controls and jump controls.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Timers, relays, counters, shift registers and controls is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: Malayalam support: lang="ml">Timers, relays, counters, shift registers and controls എന്നു് topic ഉള്ളതിനെക്കുറിച്ചു് അർത്ഥം, പ്രാധാന്യം, working sequence, variables, application, limitation, safety check എന്നിവയെക്കുറിച്ചു് വിവരണം. Standard technical terms English-ലേക്കു് മാറ്റം വരുത്തണം.

Study points

- Define Timers, relays, counters, shift registers and controls using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.

- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Timers, relays, counters, shift registers and controls is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Timers, relays, counters, shift registers and controls under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Timers, relays, counters, shift registers and controls without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **PLC data and analogue I/O**. The corresponding study scope is: Explain data handling, analogue input/output and selection of PLC.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC data and analogue I/O is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** PLC data and analogue I/O ൽ താല്പര്യമുള്ള വിഷയം പരിശോധിക്കുക. ഈ വിഷയത്തിന്റെ അർത്ഥം, ഉദ്ദേശ്യം, പ്രവർത്തനക്രമം, ഘടകങ്ങൾ, വേരിഫിക്കേഷൻ, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന എന്നിവ പരിശോധിക്കുക. സ്റ്റാൻഡർഡ് ടെക്നിക്കൽ ടേർമിംഗ് ഇംഗ്ലീഷിൽ ഉപയോഗിക്കുക.

Study points

- Define PLC data and analogue I/O using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: PLC data and analogue I/O is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for PLC data and analogue I/O under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for PLC data and analogue I/O without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **PLC and embedded-board applications**. The corresponding study scope is: Discuss timed switch, windscreen-wiper motion, bathroom scale, Arduino board, Raspberry Pi board and basic applications.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC and embedded-board applications is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** PLC and embedded-board applications topic പഠിക്കേണ്ട കാര്യങ്ങൾ meaning പഠിക്കേണ്ട purpose പഠിക്കേണ്ട steps, components പഠിക്കേണ്ട variables, പഠിക്കേണ്ട application, limitation, safety check പഠിക്കേണ്ട പഠിക്കേണ്ട പഠിക്കേണ്ട. Standard technical terms English-പഠിക്കേണ്ട പഠിക്കേണ്ട.

Study points

- Define PLC and embedded-board applications using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: PLC and embedded-board applications is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for PLC and embedded-board applications under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for PLC and embedded-board applications without explaining their order, purpose, or engineering consequence.

Module summary: A PLC program should be tested for initial state, latching, interlock, sequence, timer/counter action and safe reset.

OFFICIAL MODULE 4

Automation and robotics

The final module connects automation strategy to robot anatomy, applications, programming, design factors and case studies.

Hours: 14

CO4 · Bloom level: Applying

Learning goals

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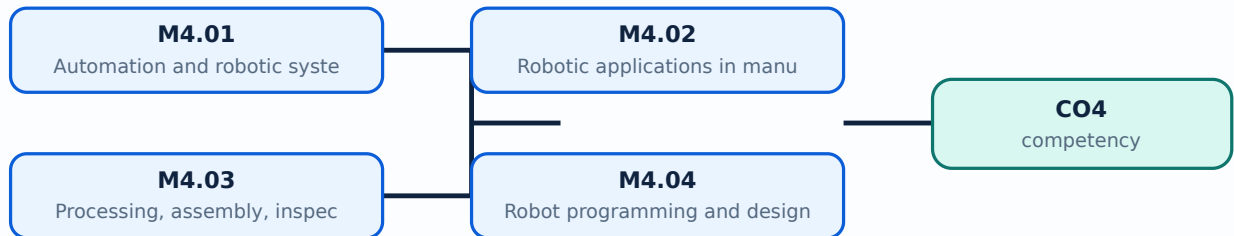
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Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

Concept and explanation

Official scope. The syllabus specifies **Automation and robotic systems**. The corresponding study scope is: Discuss automation and robotics, robotic-system anatomy, classification and future prospects.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Study frame for Automation and robotic systems	
Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Automation and robotic systems is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Automation and robotic systems എന്നു് topic ന്റെ അർത്ഥം, പ്രാധാന്യം, meaning, purpose, പ്രവർത്തനം. പ്രവർത്തനം ന്റെ steps, components, variables, application, limitation, safety check എന്നിവ ഉൾക്കൊള്ളുന്ന. Standard technical terms English-നുള്ള അർത്ഥം ഉൾക്കൊള്ളുന്ന.

Study points

- Define Automation and robotic systems using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Automation and robotic systems is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Automation and robotic systems under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Automation and robotic systems without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Robotic applications in manufacturing**. The corresponding study scope is: Identify material transfer, machine loading and unloading applications.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Robotic applications in manufacturing is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Robotic applications in manufacturing topic നോക്കുക. ഔദ്യോഗിക പദം, പ്രവർത്തന ശ്രേണി, ഘടകങ്ങൾ, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന എന്നിവയെക്കുറിച്ച് വിവരിക്കുക. ഉപയോഗം: മെക്കാട്രോണിക്സും റോബോട്ടിക്സും പ്രായോഗികത, രേഖാപ്പെടുത്തൽ, ത্রുതീകരണം, ഡിസൈൻ റിവ്യൂ, അല്ലെങ്കിൽ പരീക്ഷണ വിശ്ലേഷണം. സ്റ്റാൻഡർഡ് ടെക്നിക്കൽ പദങ്ങൾ ഇംഗ്ലീഷിൽ ഉപയോഗിക്കുക.

Study points

- Define Robotic applications in manufacturing using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Robotic applications in manufacturing is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Robotic applications in manufacturing under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Robotic applications in manufacturing without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Processing, assembly, inspection and feedback**. The corresponding study scope is: Discuss processing operations, assembly, inspection, robot activation and feedback components.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Processing, assembly, inspection and feedback is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Processing, assembly, inspection and feedback എന്നു് വിഷയം പരിചരിക്കുന്നതിനുള്ള അർത്ഥം, പ്രവർത്തനം, ഘടകങ്ങൾ, വേർതിരിവ്, പ്രയോഗം, പരിമിതി, സുരക്ഷാ പരിശോധന എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Standard technical terms English-ഇംഗ്ലീഷ് പദങ്ങൾ പരിചരിക്കുന്നതിനുള്ള അർത്ഥം.

Study points

- Define Processing, assembly, inspection and feedback using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Processing, assembly, inspection and feedback is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Processing, assembly, inspection and feedback under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Processing, assembly, inspection and feedback without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies **Robot programming and design factors**. The corresponding study scope is: Explain programming methods for robots and robot design factors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Robot programming and design factors is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Robot programming and design factors topic നോക്കുക meaning നോക്കുക purpose നോക്കുക. steps, components നോക്കുക variables, application, limitation, safety check നോക്കുക. Standard technical terms English-നോക്കുക.

Study points

- Define Robot programming and design factors using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Robot programming and design factors is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Robot programming and design factors under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Robot programming and design factors without explaining their order, purpose, or engineering consequence.

Concept and explanation

Official scope. The syllabus specifies *Mechatronics case studies*. The corresponding study scope is: Identify case studies of a pick-and-place

robot and a car-park barrier system.

How to study the topic.

Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output →**

verification

. For a design or management method, write **objective** → assumptions → method → result → limitation.</div>

Study frame for Mechatronics case studies	
Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Mechatronics case studies is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

examination analysis.

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Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Malayalam support: **Malayalam support:** Mechatronics case studies തിരഞ്ഞെടുക്കുന്ന വിഷയം തിരഞ്ഞെടുക്കുന്ന വിഷയത്തിന്റെ purpose, components, variables, application, limitation, safety check എന്നിവ തിരഞ്ഞെടുക്കുന്ന വിഷയം. Standard technical terms English-യിൽ തിരഞ്ഞെടുക്കുന്ന വിഷയം.

Study points

- Define Mechatronics case studies using the official syllabus wording.
- Explain the sequence, relationship, calculation, or document procedure before listing details.
- Use a labelled diagram, table, formula, unit, or operating condition wherever it improves the answer.
- Connect the topic to an appropriate application and state one limitation, verification point, or common error.

Engineering / workplace application: Mechatronics case studies is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Illustrative example: Revision task: Prepare a four-column note for Mechatronics case studies under the headings purpose, principle/procedure, important parameters, and application/limitation. Check every term in the official scope before moving to the next topic.

Common mistake: A common mistake is to reproduce isolated points for Mechatronics case studies without explaining their order, purpose, or engineering consequence.

Module summary: Robot selection should consider payload, reach, degrees of freedom, accuracy, speed, end effector, environment and safety.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

Module 1: Introduction to mechatronics view its labelled learning-map diagram.	Module 2: Mechanical, fluid-
Module 2: Mechanical, fluid- view its labelled learning-map diagram.	Module 3: PLC architecture
Module 3: PLC architecture view its labelled learning-map diagram.	Module 4: Automation and
Module 4: Automation and view its labelled learning-map diagram.	

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied syllabus does not provide a separate official self-learning activity list. Use the topic procedures, worked examples, diagrams, and module questions in this lesson for structured study.

Practical requirements / equipment

- The supplied syllabus does not prescribe separate practical equipment.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied PDF lists Series Test entries, but it does not provide a complete official CIE/ESE mark distribution or a complete question-paper pattern.
- Series Test - I is listed in the supplied syllabus; the PDF does not provide a complete mark distribution for this entry.
- Series Test - II is listed in the supplied syllabus; the PDF does not provide a complete mark distribution for this entry.
- The practice model paper in this lesson is a study aid and is explicitly not an official examination paper.

Module-wise Questions and Answers

module-1 — Mechatronics, control systems and sensors

1. Explain Importance and applications of Mechatronics. [3 marks]

Official scope. The syllabus specifies **Importance and applications of Mechatronics**. The corresponding study scope is: Explain importance of Mechatronics and its applications in modern industries.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition

that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Importance and applications of Mechatronics is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Control and measurement systems. [3 marks]

Official scope. The syllabus specifies *Control and measurement systems*. The corresponding study scope is: Familiarize control systems and types; open- and closed-loop control; measurement systems and measurement-system terminology.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Control and measurement systems is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical

vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Types of sensors. [3 marks]

Official scope. The syllabus specifies *Types of sensors*. The corresponding study scope is: Identify different types of sensors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Types of sensors is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Sensor selection. [3 marks]

Official scope. The syllabus specifies *Sensor selection*. The corresponding study scope is: Explain selection of sensors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Sensor selection is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Mechanical, fluid-power and electrical actuators

5. Explain Mechanical actuators and mechanisms. [3 marks]

Official scope. The syllabus specifies **Mechanical actuators and mechanisms**. The corresponding study scope is: Introduce mechanical actuators, kinematic chain, Geneva mechanism and mechanical aspects of motor selection.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Mechanical actuators and mechanisms is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Hydraulic and pneumatic systems and directional valves. [3 marks]

Official scope. The syllabus specifies **Hydraulic and pneumatic systems and directional valves**. The corresponding study scope is: Familiarize hydraulic and pneumatic systems, power-supply layout, spool/poppet/pilot-operated and directional control valves.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Hydraulic and pneumatic systems and directional valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Pressure and flow-control valves. [3 marks]

Official scope. The syllabus specifies **Pressure and flow-control valves**. The corresponding study scope is: Familiarize pressure-regulating, pressure-limiting, pressure-sequence and flow-control valves.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Pressure and flow-control valves is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Cylinders and sequencing. [3 marks]

Official scope. The syllabus specifies *Cylinders and sequencing*. The corresponding study scope is: Identify single-acting and double-acting cylinders and cylinder sequencing.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Cylinders and sequencing is applied in mechatronics systems practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — PLC architecture and programming

9. Explain PLC definition and block diagram. [3 marks]

Official scope. The syllabus specifies **PLC definition and block diagram**. The corresponding study scope is: Explain Programmable Logic Controller definition and basic block diagram.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC definition and block diagram is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain PLC input/output processing and ladder programming. [3 marks]

Official scope. The syllabus specifies **PLC input/output processing and ladder programming**. The corresponding study scope is: Discuss input/output processing; ladder diagram, logic functions, latching and sequencing.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain

the sequence, governing idea, calculation method, or document workflow.

Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC input/output processing and ladder programming is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain PLC mnemonics. [3 marks]

Official scope. The syllabus specifies **PLC mnemonics**. The corresponding study scope is: Familiarize PLC mnemonics.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	PLC mnemonics is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Timers, relays, counters, shift registers and controls. [3 marks]

Official scope. The syllabus specifies *Timers, relays, counters, shift registers and controls*. The corresponding study scope is: Discuss timers, internal relays, counters, shift registers, master controls and jump controls.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Study frame for Timers, relays, counters, shift registers and controls	
Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Timers, relays, counters, shift registers and controls is applied in plc and mechatronics control practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Automation and robotics

13. Explain Automation and robotic systems. [3 marks]

Official scope. The syllabus specifies *Automation and robotic systems*. The corresponding study scope is: Discuss automation and robotics, robotic-system anatomy, classification and future prospects.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Study frame for Automation and robotic systems	
Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	

<td>State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.</td></tr><tr><th>Application</th><td>Automation and robotic systems is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.</td></tr></tbody></table></div> <p>Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Robotic applications in manufacturing. [3 marks]

<p>Official scope. The syllabus specifies Robotic applications in manufacturing. The corresponding study scope is: Identify material transfer, machine loading and unloading applications.</p> <p>How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write input → transformation → output → verification. For a design or management method, write objective → assumptions → method → result → limitation.</p> <div class="table-wrap"> <table><caption>Study frame for Robotic applications in manufacturing</caption><thead> <tr><th>Aspect</th><th>Expected learning</th></tr></thead><tbody><tr> <th>Meaning</th><td>Define the official term and distinguish it from closely related terms. </td></tr><tr><th>Principle or procedure</th><td>Explain the sequence, governing idea, calculation method, or document workflow.</td></tr><tr><th>Engineering check</th> <td>State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.</td></tr><tr><th>Application</th><td>Robotic applications in manufacturing is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.</td></tr></tbody></table></div> <p>Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.</p>

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Processing, assembly, inspection and feedback. [3 marks]

Official scope. The syllabus specifies *Processing, assembly, inspection and feedback*. The corresponding study scope is: Discuss processing operations, assembly, inspection, robot activation and feedback components.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Processing, assembly, inspection and feedback is applied in mechatronics and robotics practice, documentation, troubleshooting, design review, or examination analysis.

Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Robot programming and design factors. [3 marks]

Official scope. The syllabus specifies *Robot programming and design factors*. The corresponding study scope is: Explain programming methods for robots and robot design factors.

How to study the topic. Begin with the exact definition or purpose, then identify the important elements, working sequence, variables or controls, and the performance or safety condition that must be checked. For a process, write **input → transformation → output → verification**. For a design or management method, write **objective → assumptions → method → result → limitation**.

Aspect	Expected learning
Meaning	Define the official term and distinguish it from closely related terms.
Principle or procedure	Explain the sequence, governing idea, calculation method, or document workflow.
Engineering check	State the parameter, accuracy, speed, cost, reliability, safety, or compliance condition to verify.
Application	Robot programming and design factors is applied in mechatronics and robotics practice,

documentation, troubleshooting, design review, or examination analysis.

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Technical treatment. Use the official scope as a checklist. Relate each item to the neighbouring topics in the module, and distinguish normal operation from a fault, limitation, or incorrect assumption. In an examination answer, use the exact technical vocabulary, a labelled diagram or table where useful, and state units or assumptions for numerical work.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. Define or explain *Importance and applications of Mechatronics* with one relevant application. (5 marks)
2. Define or explain *Control and measurement systems* with one relevant application. (5 marks)
3. Define or explain *Mechanical actuators and mechanisms* with one relevant application. (5 marks)
4. Define or explain *Hydraulic and pneumatic systems and directional valves* with one relevant application. (5 marks)
5. Define or explain *PLC definition and block diagram* with one relevant application. (5 marks)
6. Define or explain *PLC input/output processing and ladder programming* with one relevant application. (5 marks)
7. Define or explain *Automation and robotic systems* with one relevant application. (5 marks)
8. Define or explain *Robotic applications in manufacturing* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Mechatronics, control systems and sensors:** Sensor selection should match range, sensitivity, response, environment, interface and required accuracy.
- **Mechanical, fluid-power and electrical actuators:** An actuator answer should identify energy source, control element, motion/force output and feedback or safety limitation.
- **PLC architecture and programming:** A PLC program should be tested for initial state, latching, interlock, sequence, timer/counter action and safe reset.
- **Automation and robotics:** Robot selection should consider payload, reach, degrees of freedom, accuracy, speed, end effector, environment and safety.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Importance and applications of Mechatronics	<p>Official scope. The syllabus specifies Importance and applications of Mechatronics. The corresponding study scope is: Explain importance of Mechatronics and its applications in modern industries.</p> <p>How to study the top
Control and measurement systems	<p>Official scope. The syllabus specifies Control and measurement systems. The corresponding study scope is: Familiarize control systems and types; open- and closed-loop control; measurement systems and measurement-system terminology.

Term	Short meaning
Types of sensors	<p>Official scope. The syllabus specifies Types of sensors. The corresponding study scope is: Identify different types of sensors.</p> <p>How to study the topic. Begin with the exact definition or purpose, then identi
Sensor selection	<p>Official scope. The syllabus specifies Sensor selection. The corresponding study scope is: Explain selection of sensors.</p> <p>How to study the topic. Begin with the exact definition or purpose, then identify the
Sensor and controller applications	<p>Official scope. The syllabus specifies Sensor and controller applications. The corresponding study scope is: Identify sensor and controller applications in washing machines and automatic water-level controllers.</p> <p>How
Mechanical actuators and mechanisms	<p>Official scope. The syllabus specifies Mechanical actuators and mechanisms. The corresponding study scope is: Introduce mechanical actuators, kinematic chain, Geneva mechanism and mechanical aspects of motor selection.</p> <p><stro
Hydraulic and pneumatic systems and directional valves	<p>Official scope. The syllabus specifies Hydraulic and pneumatic systems and directional valves. The corresponding study scope is: Familiarize hydraulic and pneumatic systems, power-supply layout, spool/poppet/pilot-operated and dire
Pressure and flow-control valves	<p>Official scope. The syllabus specifies Pressure and flow-control valves. The corresponding study scope is: Familiarize pressure-regulating, pressure-limiting, pressure-sequence and flow-control valves.</p> <p>How to study t
Cylinders and sequencing	<p>Official scope. The syllabus specifies Cylinders and sequencing. The corresponding study scope is: Identify single-acting and double-acting cylinders and cylinder sequencing.</p> <p>How to study the topic. Begin wi
Process and rotary actuators	<p>Official scope. The syllabus specifies Process and rotary actuators. The corresponding study scope is: Familiarize process-control valve, diaphragm actuator, rotary actuator and semi-rotary actuator.</p> <p>How to study the
Electrical actuation systems	<p>Official scope. The syllabus specifies Electrical actuation systems. The corresponding study scope is: Describe mechanical switches, relays, solid-state switches, diodes, thyristors, triacs, bipolar transistors and solenoids.</p> <p><
Stepper motors	<p>Official scope. The syllabus specifies Stepper motors. The corresponding study scope is: Explain the working principle of stepper motors.</p> <p>How to study the topic. Begin with the exact definition or purpose, t
Basic lift operation	<p>Official scope. The syllabus specifies Basic lift operation. The corresponding study scope is: Discuss the application of basic lift operation.</p> <p>How to study the topic. Begin with the exact definition or purp

Term	Short meaning
PLC definition and block diagram	<p>Official scope. The syllabus specifies PLC definition and block diagram. The corresponding study scope is: Explain Programmable Logic Controller definition and basic block diagram.</p> <p>How to study the topic. Be
PLC input/output processing and ladder programming	<p>Official scope. The syllabus specifies PLC input/output processing and ladder programming. The corresponding study scope is: Discuss input/output processing; ladder diagram, logic functions, latching and sequencing.</p> <p>
PLC mnemonics	<p>Official scope. The syllabus specifies PLC mnemonics. The corresponding study scope is: Familiarize PLC mnemonics.</p> <p>How to study the topic. Begin with the exact definition or purpose, then identify the import
Timers, relays, counters, shift registers and controls	<p>Official scope. The syllabus specifies Timers, relays, counters, shift registers and controls. The corresponding study scope is: Discuss timers, internal relays, counters, shift registers, master controls and jump controls.</p> <p>
PLC data and analogue I/O	<p>Official scope. The syllabus specifies PLC data and analogue I/O. The corresponding study scope is: Explain data handling, analogue input/output and selection of PLC.</p> <p>How to study the topic. Begin with the e
PLC and embedded-board applications	<p>Official scope. The syllabus specifies PLC and embedded-board applications. The corresponding study scope is: Discuss timed switch, windscreen-wiper motion, bathroom scale, Arduino board, Raspberry Pi board and basic applications.<
Automation and robotic systems	<p>Official scope. The syllabus specifies Automation and robotic systems. The corresponding study scope is: Discuss automation and robotics, robotic-system anatomy, classification and future prospects.</p> <p>How to study the
Robotic applications in manufacturing	<p>Official scope. The syllabus specifies Robotic applications in manufacturing. The corresponding study scope is: Identify material transfer, machine loading and unloading applications.</p> <p>How to study the topic.
Processing, assembly, inspection and feedback	<p>Official scope. The syllabus specifies Processing, assembly, inspection and feedback. The corresponding study scope is: Discuss processing operations, assembly, inspection, robot activation and feedback components.</p> <p>H
Robot programming and design factors	<p>Official scope. The syllabus specifies Robot programming and design factors. The corresponding study scope is: Explain programming methods for robots and robot design factors.</p> <p>How to study the topic. Begin w
Mechatronics case studies	<p>Official scope. The syllabus specifies Mechatronics case studies. The corresponding study scope is: Identify case studies of a pick-and-place robot and a car-park barrier system.</p> <p>How to study the topic. Begi

Official References and Resources

References listed in the supplied syllabus

1. W. Bolton, Mechatronics, Pearson Education India.
2. R.K. Rajput, A Text Book on Mechatronics, S. Chand & Co.
3. M.D. Singh and Joshi, Mechatronics, Prentice Hall of India.
4. HMT, Mechatronics, Tata McGraw-Hill.
5. Devadas Shetty, Mechatronics System, PWS Publishing.
6. Pradeep Kumar Srivatsava, Exploring Programmable Logic Controllers with Applications, BPB Publications.

Official learning resources listed

- <https://youtu.be/zVVITxiec7g>
- <https://youtu.be/8Qiy2nf4sp0>
- <https://youtu.be/XI2nWDcy0To>
- <https://nptel.ac.in/course.html>

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